

SHREYA UDAY HASYAGAR

Senior Software Engineer - Python · AWS · Distributed Data Systems · Applied AI

Bengaluru, India · shreya.uday.hasyagar@gmail.com · linkedin.com/in/shreya-hasyagar · github.com/shreyahasyagar

PROFESSIONAL SUMMARY

Senior Software Engineer with 4+ years building production, event-driven data-enrichment platforms on AWS. Specializes in distributed serverless pipelines, entity resolution, and applied AI / LLM systems in Python. Owns work end to end - system design, reliability engineering, and AI-assisted automation across multi-tenant microservices. Experienced working in a remote-first, async environment.

SKILLS

Languages: Python · SQL · JavaScript / TypeScript · Bash

Backend & APIs: FastAPI · Flask · Django REST · SQLAlchemy · asyncio · REST design (idempotency, pagination, auth)

Applied AI / LLM: OpenAI GPT-4 / 4o · Anthropic Claude · open models (Llama) · prompt engineering · RAG · vector databases · evaluation harnesses · MCP · LangGraph agents

Cloud (AWS): Lambda · Step Functions · SQS · SNS · DynamoDB · RDS · S3 · API Gateway · ECS / Fargate · Kinesis Firehose · IAM

Data & Search: PostgreSQL · Snowflake · OpenSearch · Redis / ElastiCache

Architecture: Distributed systems · Event-driven · Microservices · Caching · System design (LLD & HLD) · Concurrency

DevOps & IaC: Docker · Terraform · AWS CDK · GitLab CI/CD · Snyk · Git

Observability: New Relic · CloudWatch · OpenTelemetry · structured logging · Slack alerting

WORK EXPERIENCE

Senior Software Engineer - Cimpres India (Remote-first)

01/2025 – Present

- Primary author of a ~30-stage serverless AWS Step Functions enrichment pipeline integrating 8 third-party data providers, with exponential-backoff retries and a Firehose → S3 → Snowflake sink; each provider stage is independently deployable.
- Re-architected the entity-resolution identity graph from DynamoDB to Amazon RDS (PostgreSQL), replacing check-then-skip logic with an atomic insert-or-retry concurrency model - eliminating duplicate/orphan records across 10+ teams and 100K+ entities (100% pass on the race-limit test harness).
- Built an LLM-powered document-classification API, cutting classification latency 50% and manual effort 85% at under \$0.01 per document (5K+ documents/month).
- Cut manual pipeline intervention 80% with an automated failed-execution recovery system (EventBridge → DynamoDB → scheduled Fargate re-runs) with a polling Lambda and Slack alerting.
- Raised reliability across 25+ microservices via 2,300+ code-review comments enforcing layered exception handling, least-privilege IAM, and type-safety contracts as merge gates - blocking high-severity defects (XSS, SSRF, ReDoS).
- Automated secret rotation across 20+ repositories via a Secrets Manager-triggered Lambda (100% rotation-policy compliance); led remediation of 30+ Snyk security findings.

Software Engineer - Cimpres India (Remote-first)

08/2022 – 12/2024

- Re-engineered a search API from 30s+ timeouts to under 5s by optimizing PostgreSQL full-text search (ts_query, ts_vector, ts_rank) and fixing a multi-word query-parsing bug.
- Improved API performance 45% and reduced provider cost with a TTL-based DynamoDB caching layer across multiple regional pipelines.
- Lifted industry-classification accuracy 40% with AI-assisted taxonomy mapping, and raised enrichment coverage from 41% to over 60% across 4,000+ records.
- Established performance benchmarks with Apache JMeter (stable to 2,500 concurrent users, ~41 req/s, ~1.2s latency); right-sized Lambda memory via power tuning.
- Reduced CI/CD failures 25% with 15+ test alert hooks and tuned GitLab runners; delivered secure, incremental releases including Snyk fixes and GDPR compliance workflows.

Programmer Analyst Intern - Cognizant Technology Solutions

02/2022 – 07/2022

- Reduced IAM deployment time 25% by configuring directory schemas and automating admin-client setup for 3 access workflows.
- Developed 30+ unit tests for identity modules, cutting QA iterations 20% and speeding feedback cycles.

PROJECTS

Company Enrichment API & Evaluation Harness

- Python FastAPI + PostgreSQL service with a Redis cache, retry logic, and a dead-letter table; an evaluation harness computes per-field precision/recall behind a CI accuracy gate.

MCP Server & Retrieval-Augmented AI Agent

- Model Context Protocol server exposing tool-connected AI agents over public APIs, plus a LangGraph RAG agent over a vector database with authorization guardrails and an evaluation suite.

Underwater Image Enhancement - Conditional GAN

- Final-year research: a conditional GAN in Python & TensorFlow that restores clarity and corrects color in underwater imagery, validated with quantitative image-quality metrics (PSNR/SSIM) and visual inspection.

Personal Portfolio & Résumé Site

- Hand-built, zero-framework portfolio (HTML/CSS/JS) with a persisted light/dark theme, animated hero, and print-optimized résumé; deployed on GitHub Pages.

EDUCATION

B.E., Computer Science Engineering - 2022

RNS Institute of Technology, Bengaluru · GPA 3.75 / 4.00

CERTIFICATIONS

AWS Certified Developer – Associate

AWS Certified Cloud Practitioner

Prompt Engineering - Vanderbilt University